IT PROJECT MANAGEMENT —TUTORIAL

Critical Path Method (CPM)

Learning Outcomes

- Build a network diagram from a list of activities and their dependencies
- Perform a forward pass to calculate Early Start (ES) and Early Finish (EF) for every activity
- Perform a backward pass to calculate Late Start (LS) and Late Finish (LF) for every activity
- Calculate total float and identify every critical activity
- Determine the critical path (or paths) and state the project duration
- Judge whether a delay to a specific activity will delay the whole project — or not

1. Key Concepts

What Is the Critical Path?

Every project schedule is really a network of activities, some of which can happen in parallel and some of which must happen in strict sequence. The critical path is the single longest chain of dependent activities running from the start of the project to the end — and its total duration is the shortest possible time the project can be completed in.

Knowing the critical path tells a project manager exactly where schedule risk lives: a delay on a critical activity delays the whole project, day for day, while a delay on a non-critical activity may not matter at all.

Key Terms

Term	Meaning
Activity	A piece of work with a defined duration (e.g. 'Backend Development — 6 days').
Predecessor	An activity that must be completed (or started) before another activity can begin.
Network Diagram	A diagram of boxes (activities) and arrows (dependencies) showing the order of work.
Path	Any complete sequence of connected activities from the project start to the project end.
Critical Path	The longest path through the network — it sets the shortest possible project duration.
Float (Slack)	How many days an activity can slip without delaying the overall project finish date.
Critical Activity	Any activity with zero float — a delay here delays the whole project.

Key insight: the critical path isn't a list of the 'most important' activities in a subjective sense — it is simply the longest path, found by arithmetic. A small-looking task can turn out to be critical, and a seemingly major task can have days of slack.

2. The 4-Step Calculation Method

Step 1 — Draw the Network Diagram

List every activity together with its duration and its predecessor(s), then draw it as boxes connected left to right by arrows, in the order that dependencies allow.

Step 2 — Forward Pass: Calculate ES and EF

ES (first activity) = 0

EF = ES + Duration

ES (any later activity) = MAX(EF of ALL its predecessors)

Work left to right through the diagram. When an activity has more than one predecessor, take the LARGEST (latest) EF among them — the activity cannot start until every one of its predecessors has finished.

Step 3 — Backward Pass: Calculate LS and LF

LF (last activity) = Project Duration (the largest EF found above)

LS = LF − Duration

LF (any earlier activity) = MIN(LS of ALL its successors)

Work right to left through the diagram. When an activity feeds more than one successor, take the SMALLEST (earliest) LS among them — it must finish in time for the earliest-starting successor.

Step 4 — Calculate Float and Identify the Critical Path

Total Float = LS − ES (this always equals LF − EF too)

Critical Path = every activity where Float = 0

Trace the zero-float activities from start to finish — that chain is the critical path, and its total duration equals the project duration. It is entirely possible for two or more chains to tie for the longest duration, giving a project more than one critical path.

3. Worked Example

Scenario: Mobile App Release

Activity	Predecessor(s)	Duration (days)
A — Requirements	Start	3
B — UI Design	A	4
C — Backend Development	A	6
D — Integration	B, C	2
E — Testing	D	3

Forward Pass (left to right)

- A: no predecessor, so ES = 0. EF = 0 + 3 = 3.
- B: predecessor A finishes at 3, so ES = 3. EF = 3 + 4 = 7.
- C: predecessor A finishes at 3, so ES = 3. EF = 3 + 6 = 9.
- D: predecessors are B (EF=7) and C (EF=9) — take the larger: ES = 9. EF = 9 + 2 = 11.
- E: predecessor D finishes at 11, so ES = 11. EF = 11 + 3 = 14.

→ Project duration = 14 days (the largest EF, from activity E).

Backward Pass (right to left)

- E: last activity, so LF = 14 (the project duration). LS = 14 − 3 = 11.
- D: only successor is E (LS=11), so LF = 11. LS = 11 − 2 = 9.
- C: only successor is D (LS=9), so LF = 9. LS = 9 − 6 = 3.
- B: only successor is D (LS=9), so LF = 9. LS = 9 − 4 = 5.
- A: successors are B (LS=5) and C (LS=3) — take the smaller: LF = 3. LS = 3 − 3 = 0.

Float and Critical Path

Activity	ES	EF	LS	LF	Float	Critical?
A	0	3	0	3	0	Yes
B	3	7	5	9	2	No
C	3	9	3	9	0	Yes
D	9	11	9	11	0	Yes
E	11	14	11	14	0	Yes

Critical Path: A → C → D → E = 3 + 6 + 2 + 3 = 14 days (matches the project duration ✓). Activity B has 2 days of float — it could slip by up to 2 days without affecting the project finish date.

EXERCISE 1

Basic Network — One Merge Point

Calculate ES, EF, LS, LF and Float for every activity, then state the critical path and project duration.

Activity	Predecessor(s)	Duration (days)
A	Start	4
B	Start	2
C	A, B	5
D	C	3

Show Your Working

Summary Table

Activity	ES	EF	LS	LF	Float	Critical?
A						
B						
C						
D						

Critical Path:

Project Duration:

EXERCISE 2

Medium Network — Tied Critical Paths

Calculate ES, EF, LS, LF and Float for every activity. Watch out — this network may have more than one critical path.

Activity	Predecessor(s)	Duration (days)
A	Start	5
B	Start	3
C	A	4
D	B	6
E	B	2
F	C, D, E	3

Show Your Working

Summary Table

Activity	ES	EF	LS	LF	Float	Critical?
A						
B						
C						
D						
E						
F						

Critical Path(s):

Project Duration:

EXERCISE 3

Applying Float — Interpreting Delays

Calculate the full table as before, then use your Float column to answer the interpretation questions — this is the real-world skill the whole technique is for.

Activity	Predecessor(s)	Duration (days)
A	Start	6
B	Start	4
C	A	3
D	B	8
E	C, D	4

Show Your Working

Summary Table

Activity	ES	EF	LS	LF	Float	Critical?
A						
B						
C						
D						
E						

Critical Path:

Interpretation Questions

1. If activity D slips by 3 days, does the project finish date change? By how much, and why?

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2. If activity C slips by 2 days, does the project finish date change? Why or why not?

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3. If activity C slips by 5 days instead, does the project finish date change? By how much?